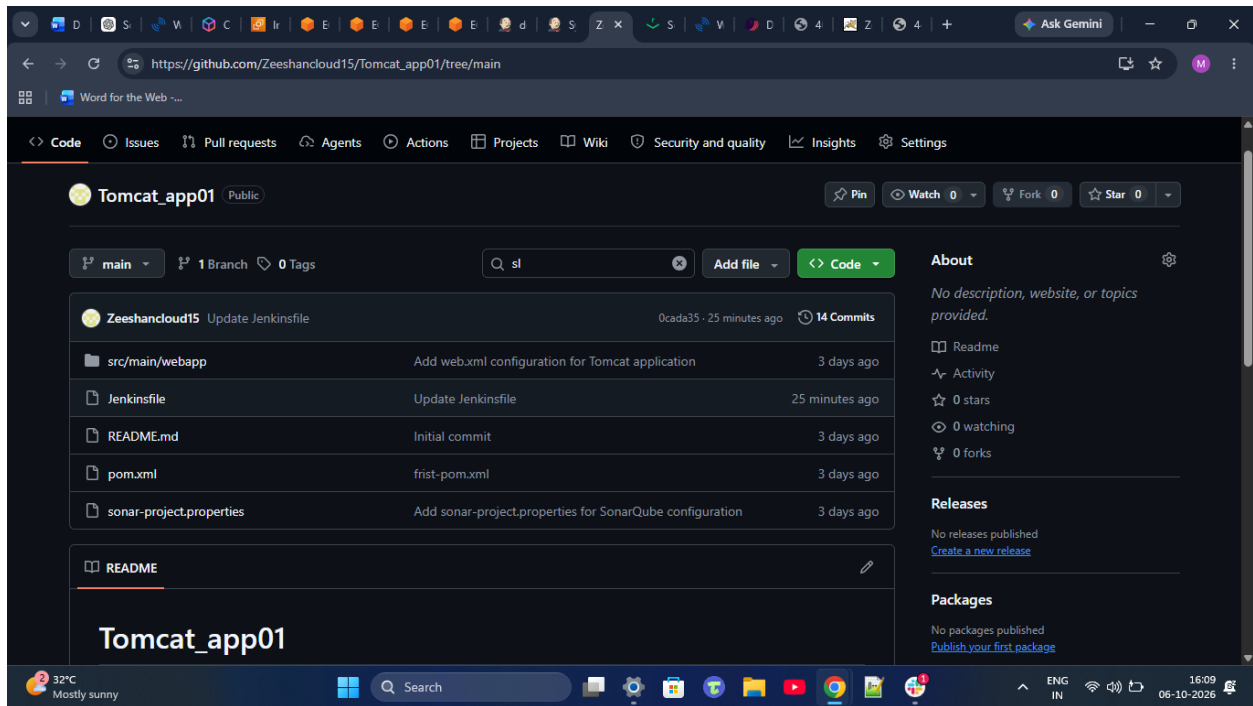


EOD – Jenkins CI/CD Pipeline

1. Project Setup

1. Created Java Maven web application.
2. Configured the project as a WAR application.
3. Created GitHub repository.
4. Pushed the project to the main branch.
5. Added the following files:
 - a. pom.xml
 - b. Jenkinsfile
 - c. sonar-project.properties
 - d. README.md
 - e. Src/main/webapp/



2. GitHub Configuration

1. Created the GitHub repository:
Tomcat_app01
2. Used the main branch.
3. Added the Jenkinsfile to the repository.
4. Verified that Jenkinsfile, POM, SonarQube configuration and application source code were available in GitHub.

3. Jenkins Job Configuration

1. Created a Jenkins Pipeline job.

2. Configured the pipeline to obtain the Jenkinsfile from SCM.
3. Selected Git as the SCM.
4. Added the GitHub repository.
5. Selected the main branch.
6. Configured Jenkins to execute the Jenkinsfile from the repository.

4. Maven Configuration

1. Opened Jenkins.
2. Navigated to:

Manage Jenkins → Tools

3. Configured Maven installation.
4. Maven installation name:

maven-integration

5. Used the Maven installation in the Jenkinsfile.

5. Jenkins Credentials

Configured the required credentials in Jenkins.

Nexus Credential

Credential ID: nexus-creid

Used for uploading the WAR artifact to Nexus.

Tomcat Credential

Credential ID: tomcat-jenkins

Used for deploying the WAR to Tomcat Manager.

6. Stage 1 – Checkout

1. Jenkins starts the pipeline.
2. Jenkins connects to GitHub.
3. Jenkins checks out the main branch.
4. Source code is downloaded into the Jenkins workspace.

Git repository:

https://github.com/Zeeshancloud15/Tomcat_app01.git

7. Stage 2 – Maven Build

1. Jenkins enters the Build stage.
2. Maven is loaded using the configured Jenkins Maven installation.
3. Executed:

mvn clean package

4. Previous build files are removed.
5. Application source code is compiled.
6. Maven packages the application as a WAR.

7. Generated WAR:

target/tomcat-app01.war

8. Stage 3 – WAR Verification

1. Listed the target directory.
2. Verified that the WAR file was generated.
3. Used:

```
ls -lh target/
```

4. Verified the file:

```
test -f target/tomcat-app01.war
```

5. Pipeline continues only when the WAR exists.

9. Stage 4 – SonarQube Analysis

1. Configured SonarQube server in Jenkins.
2. Jenkins SonarQube server name:

```
sonar-server
```

3. Configured SonarScanner:

```
/opt/sonar-scanner/bin/sonar-scanner
```

4. Added sonar-project.properties.
5. Started SonarQube analysis from Jenkins.
6. SonarQube analyzed the project source code.

10. Stage 5 – SonarQube Quality Gate

1. Jenkins waits for the SonarQube Quality Gate result.
2. Configured a 1-minute timeout.
3. Used:

```
waitForQualityGate
```

4. SonarQube sends the result back to Jenkins using the configured webhook.
5. Quality Gate result is checked before continuing the pipeline.

11. SonarQube Webhook

Configured the SonarQube webhook to communicate with Jenkins.

Webhook:

<http://3.235.2.67:8080/sonarqube-webhook/>

Flow:

Jenkins



SonarQube Analysis



Quality Gate

↓
SonarQube Webhook
↓
Jenkins

12. Stage 6 – Upload WAR to Nexus

1. After the SonarQube stage, Jenkins uploads the WAR to Nexus.
2. Nexus server:
<http://3.95.22.37:8081>
3. Repository:
maven-release
4. Group ID:
com.zeeshan
5. Artifact ID:
tomcat-app01
6. Version:
1.0
7. Artifact type:
war
8. WAR uploaded:
target/tomcat-app01.war

13. Stage 7 – Tomcat Deployment

1. Configured the Tomcat server.
2. Tomcat server:
35.169.107.79
3. Tomcat port:
8080
4. Tomcat URL:
<http://35.169.107.79:8080>
5. Configured Tomcat Manager user with:
manager-script
6. Configured Jenkins credential:
tomcat-jenkins
7. Configured deployment context:
/declarativejob
8. Jenkins deploys:
target/tomcat-app01.war
9. Final application URL:
<http://35.169.107.79:8080/declarativejob/>

14. Deployment Issue – 401 Unauthorized

During deployment, Jenkins initially reported:

HTTP request failed, response code: 401 Unauthorized

URI: /manager/text/list

Steps Taken

1. Checked the Tomcat URL configured in Jenkins.
2. Found that Jenkins was pointing to:

<http://3.235.2.67:8080>

3. Identified that this was the Jenkins server, not Tomcat.
4. Checked the actual Tomcat server.
5. Correct Tomcat URL:

<http://35.169.107.79:8080>

6. Updated the Jenkinsfile.
7. Ran the pipeline again.

15. Deployment Issue – 403 Forbidden

After correcting the Tomcat URL, Jenkins reported:

HTTP request failed, response code: 403

URI: /manager/text/list

Steps Taken

1. Confirmed that Jenkins was now reaching the actual Tomcat server.
2. Checked Tomcat Manager configuration.
3. Opened:

webapps/manager/META-INF/context.xml

4. Found:

```
<Valve className="org.apache.catalina.valves.RemoteCIDRValve"
    allow="127.0.0.0/8,:::1/128,172.31.2.13" />
```

5. Identified that the RemoteCIDRValve was restricting remote access to Tomcat Manager.
6. Jenkins server was not included in the allowed IP list.

16. Resolution – Tomcat Manager Access Restriction

For the development/testing environment, the restriction was removed.

Removed:

```
<Valve className="org.apache.catalina.valves.RemoteCIDRValve"
    allow="127.0.0.0/8,:::1/128,172.31.2.13" />
```

Updated context.xml.

Then restarted Tomcat:

```
sudo systemctl restart tomcat
Checked Tomcat:
sudo systemctl status tomcat
Expected:
Active: active (running)
```

17. Tomcat Manager API Testing

Tested Tomcat Manager from the Jenkins server.

Command:

```
curl -i -u 'jenkins:Jenkins@123' \
http://35.169.107.79:8080/manager/text/list
```

The API tested:

/manager/text/list

This verified Tomcat Manager connectivity and authentication before running the Jenkins deployment again.

18. Slack Integration

Slack notification was added to the Jenkins pipeline.

Steps

1. Created/selected the Slack notification channel.
2. Installed/configured the Jenkins Slack Notification plugin.
3. Configured Slack integration in Jenkins.
4. Configured Slack authentication/credentials where required.
5. Added slackSend to the Jenkinsfile.
6. Added a success notification.
7. Added a failure notification.

19. Slack Success Notification

When the complete pipeline succeeds, Jenkins sends a Slack notification containing:

Job name

Build number

Build status

Application URL

Jenkins Build URL

Example:

SUCCESS: Jenkins Pipeline

Job: decrartive

Build: #11

Status: SUCCESS

Application:

<http://35.169.107.79:8080/declarativejob/>

Build URL:

Jenkins build URL

20. Slack Failure Notification

If any pipeline stage fails, Jenkins sends:

FAILED: Jenkins Pipeline

Job: decrartive

Build: #11

Status: FAILED

Build URL:

Jenkins build URL

This allows the team to identify pipeline failures through Slack.

21. Final Jenkinsfile

```
pipeline {  
  
    agent any  
  
    tools {  
        maven 'maven-integration'  
    }  
  
    environment {  
        APP_NAME = 'tomcat-app01'  
        CONTEXT_PATH = '/declarativejob'  
  
        NEXUS_URL = 'http://3.95.22.37:8081'  
        NEXUS_CREDENTIALS = 'nexus-creid'  
  
        TOMCAT_URL = 'http://35.169.107.79:8080'  
    }  
  
    stages {
```

```

stage('Checkout') {
    steps{

        git branch: 'main',
            url: 'https://github.com/Zeeshancloud15/Tomcat_app01.git'
    }
}

stage('Build') {
    steps {
        sh 'mvn clean package'
    }
}

stage('Verify WAR') {
    steps {
        sh '''
            ls -lh target/
            test -f target/tomcat-app01.war
        '''
    }
}

stage('SonarQube Analysis') {
    steps {
        withSonarQubeEnv('sonar-server') {
            sh '''
                /opt/sonar-scanner/bin/sonar-scanner
            '''
        }
    }
}

stage('Quality Gate') {
    steps {
        timeout(time: 1, unit: 'MINUTES') {
            waitForQualityGate abortPipeline: false
        }
    }
}

stage('Upload to Nexus') {
    steps {

```



```

nexusArtifactUploader(
    nexusVersion: 'nexus3',
    protocol: 'http',
    nexusUrl: '3.95.22.37:8081',
    groupId: 'com.zeeshan',
    version: '1.0',
    repository: 'maven-release',
    credentialsId: 'nexus-creid',
    artifacts: [
        [
            artifactId: 'tomcat-app01',
            classifier: '',
            file: 'target/tomcat-app01.war',
            type: 'war'
        ]
    ]
)
}
}

stage('Deploy to Tomcat') {
    steps {
        deploy adapters: [
            tomcat9(
                credentialsId: 'tomcat-jenkins',
                path: '',
                url: "${TOMCAT_URL}"
            )
        ],
        contextPath: "${CONTEXT_PATH}",
        war: 'target/tomcat-app01.war'
    }
}

post {
    success {
        echo 'Pipeline completed successfully.'

        echo "Application: ${TOMCAT_URL}${CONTEXT_PATH}/"

        slackSend(
            color: 'good',

```

```
        message: ""
SUCCESS: Jenkins Pipeline
```

```
Job: ${env.JOB_NAME}
Build: #${env.BUILD_NUMBER}
Status: SUCCESS
```

```
Application:
${TOMCAT_URL}${CONTEXT_PATH}/
```

```
Build URL:
${env.BUILD_URL}
""
```

```
    )
}

failure {
    echo 'Pipeline failed.'
```

```
        slackSend(
            color: 'danger',
            message: ""
FAILED: Jenkins Pipeline
```

```
Job: ${env.JOB_NAME}
Build: #${env.BUILD_NUMBER}
Status: FAILED
```

```
Build URL:
${env.BUILD_URL}
""
```

```
    )
}

always {
    echo 'Pipeline execution completed.'
}
}
}
```

Browser window showing SonarQube community interface. URL: <http://3.235.160.82:9000/projects>

Notifications:

- Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)
- There's a new version of SonarQube Server available. Upgrade to SonarQube Server and get access to enterprise features. [Learn More](#)
- The way in which security, reliability, and maintainability counts and ratings are calculated has changed. [Learn more in SonarQube documentation](#)

SonarQube community navigation: Projects, Issues, Rules, Quality profiles, Quality gates, Administration, More

Filters:

- Quality gate: ☒ Passed (1), ☒ Failed (0)
- Security: ☒ ≥ 0 info issues (1), ☒ ≥ 1 low issue (0), ☒ ≥ 1 medium issue (0)

Project details for Tomcat App01 (Public):

- Last analysis: 15 minutes ago • 21 Lines of Code • JSP, XML
- Overall Status: **Passed**
- Metrics: Security (A 0), Reliability (C 1), Maintainability (A 0), Hotspots Reviewed (A —), Coverage (—), Duplications (0.0%)

1 of 1 shown

Browser window showing Sonatype Nexus repository interface. URL: <http://3.95.22.37:8081/#browse/browse:maven-release:com%2Fzeeshan%2Ftomcat-app01>

Sonatype nexus repository Community Edition

Search components or CVEs...

Navigation: Browse / maven-release

Upload component | HTML View | Advanced search...

Folder structure:

- com
 - zeeshan
 - tomcat-app01
 - 1.0
 - tomcat-app01-1.0.war
 - tomcat-app01-1.0.war.md5
 - tomcat-app01-1.0.war.sha1

com/zeeshan/tomcat-app01

Delete folder

```
-rw-r--r--. 1 jenkins jenkins 3277 Oct 6 10:13 jenkinsfile
-rw-r--r--. 1 jenkins jenkins 14 Oct 6 07:39 README.md
-rw-r--r--. 1 jenkins jenkins 972 Oct 6 07:39 pom.xml
-rw-r--r--. 1 jenkins jenkins 131 Oct 6 07:39 sonar-project.properties
drwxr-xr-x. 3 jenkins jenkins 18 Oct 6 07:39 src
drwxr-xr-x. 4 jenkins jenkins 72 Oct 6 10:30 target
[root@ip-172-31-2-13 dectrartive]# cd target
[root@ip-172-31-2-13 target]# ls
maven-archiver tomcat-app01 tomcat-app01.war
[root@ip-172-31-2-13 target]# pwd
/var/lib/jenkins/workspace/dectrartive/target
[root@ip-172-31-2-13 target]# ;10~
```



jenkins APP 16:00

SUCCESS: Jenkins Pipeline

Job: dectrartive

Build: #13

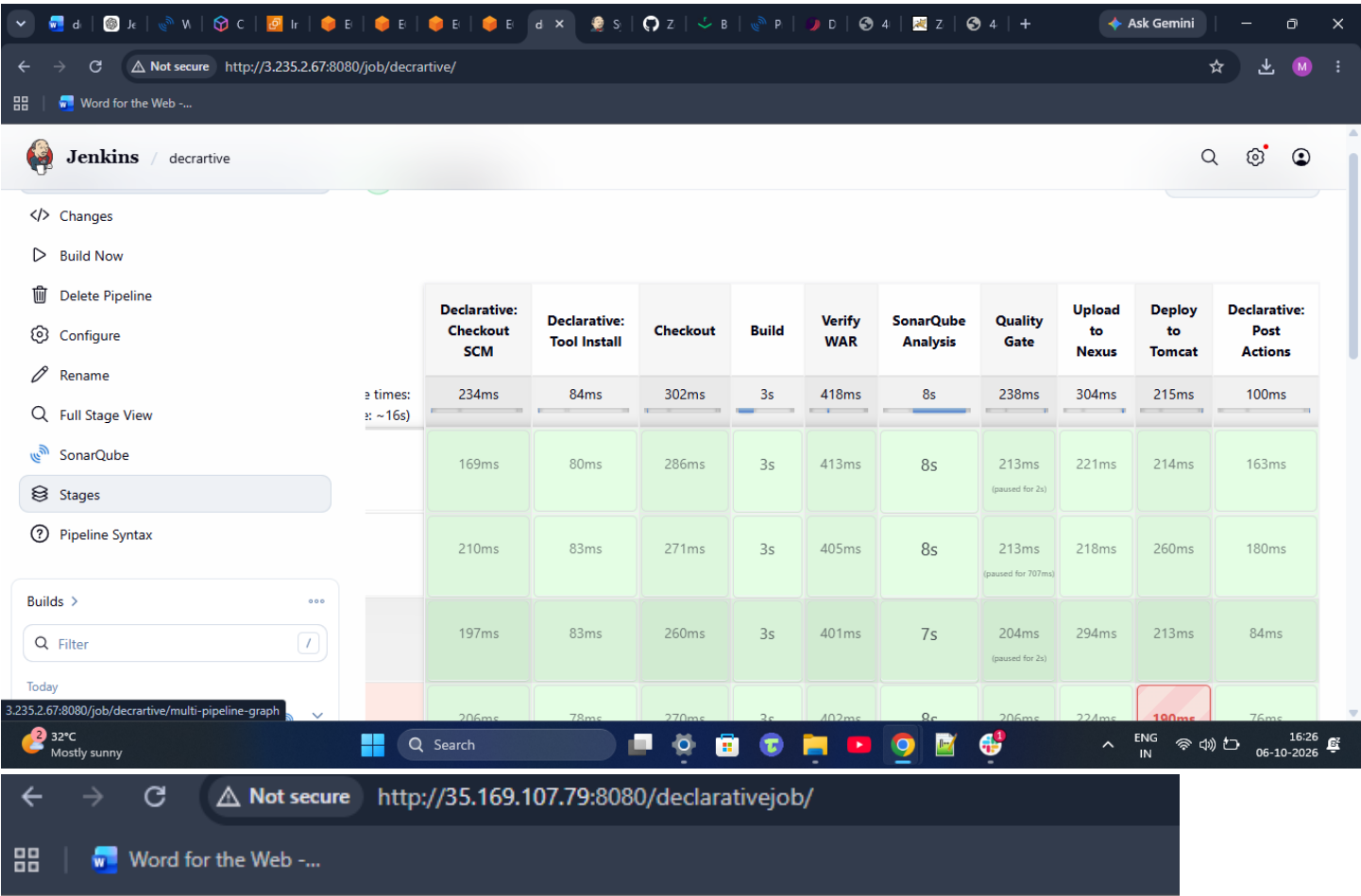
Status: SUCCESS

Application:

<http://35.169.107.79:8080/declarativejob/>

Build URL:

<http://44.195.88.168:8080/job/dectrartive/13/>



Welcome to Zeeshan Cloud Tech

Jenkins CI/CD Deployment Successful!

Application: Tomcat App01

GitHub → Jenkins → SonarQube → Maven → Nexus → Slack → Tomcat

EOD Outcome

Completed: Jenkins CI/CD pipeline implementation and deployment troubleshooting.
Pipeline: GitHub → Jenkins → Maven → SonarQube → Quality Gate → Nexus → Tomcat → Slack.
Major deployment issue resolved: Tomcat Manager 403 Forbidden caused by RemoteCIDRValve IP restriction. The restriction was removed for the development/testing environment and Tomcat was restarted.
::